

KARANN MISHRA

Python Software Engineer | Backend | AI/ML

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PROFILE

Python-focused software engineer with hands-on experience developing backend-oriented applications, database-driven systems, and AI/ML workflows. Built and evolved **AIMS/FCOS** and **ALAMS** through AURXON, delivered independent Python NLP/ML engineering work for a France-based FinTech client, and developed Django applications. Strongest areas include Python, REST APIs, SQL/PostgreSQL, Django, Git, debugging, data processing, NLP, and machine learning. Known for rapidly learning unfamiliar technical domains and turning requirements into working software through iterative analysis, implementation, testing, and refinement.

TECHNICAL SKILLS

Languages & Programming: Python, Object-Oriented Programming, Data Structures, Java, C++/C

Backend & APIs: Django, Flask, REST APIs, Web Services, API Integration

Databases: SQL, PostgreSQL, MySQL, Relational Database Concepts

Engineering Tools: Git, GitHub, Linux, Software Documentation, Problem Solving

Data & AI/ML: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, NLP, FinBERT, MiniLM

PROFESSIONAL EXPERIENCE

AURXON

Software Engineer

Aug 2024 – Present | Part-time Engagement

Indore / Remote

- Develop and iterate on software systems spanning backend logic, databases, APIs, authentication, role-based access control, debugging, documentation, and AI/ML functionality.
- Designed and evolved **AIMS (Aurxon Intelligent Management System)** into **FCOS (Founder & Company Operating System)**, a modular company-operations platform covering workforce, projects, HR, documents, authentication, and operational workflows.
- Architected and developed **ALAMS (Aurxon Lab Access Management System)**, integrating backend services, PostgreSQL-backed workflows, REST/WebSocket communication, authentication, device management, and controlled workstation access.
- Analyze requirements, design implementation approaches, troubleshoot technical issues, refactor components, validate behavior, and maintain technical documentation across software projects.

Independent Freelance Engagement

NLP & ML Engineer — France-based FinTech Client

Nov 2024 – Jan 2026

- Developed Python workflows for processing SEC regulatory filings, including document extraction, cleaning, preprocessing, and structured analysis of Risk Factors and MD&A sections.
- Implemented **FinBERT** and **MiniLM / Sentence Transformers** for sentiment analysis, semantic similarity, and financial-text analysis.
- Built validation, fallback, preprocessing, and scoring workflows for large collections of financial documents and downstream analytical research.

SUAS, Indore

Trainer of Practice — SCSIT Department

Sep 2025 – Present

Indore, India

- Conduct hands-on practical sessions in programming, AI/ML, debugging, and project-based implementation for undergraduate Computer Science students.
- Mentor technical projects and guide students in translating programming and computing concepts into working implementations.

GeekTheory Pvt. Ltd.

Project Intern — AI/ML

Mar 2024 – Jul 2024

Ujjain, India

- Contributed to machine-learning application development, application integration, custom mobile plugins, prompt-engineering workflows, and technical documentation.

SELECTED ENGINEERING PROJECTS

AIMS / FCOS — Company Operations Platform

Software Platform / Database / APIs / RBAC

- Designed a modular operating platform to centralize workforce, project, HR, document, authentication, and company-operations workflows.
- Worked across database-driven application logic, API workflows, authentication, role-based access control, notifications, and system-level integration.
- Iterated on architecture and production-readiness through debugging, validation, security review, data verification, and progressive system hardening.

ALAMS — Aurxon Lab Access Management System

Backend / PostgreSQL / REST/WebSockets / Systems

- Built a lab access-management system connecting workstation software with backend and database services for controlled authentication and access workflows.
- Implemented and analyzed REST/WebSocket communication, PostgreSQL-backed workflows, device identification, access control, and operational management components.
- Investigated workstation-locking bypass scenarios and worked through production-hardening requirements for a controlled Windows-lab environment.

Python Quiz Application

Python / Django / SQL

- Developed a Django-based application supporting faculty and student workflows, quiz creation, authentication, attempts, and progress reporting.
- Implemented role-based access for faculty and students and connected application workflows with persistent database-backed data.

SEC Financial NLP Processing Engine

Python / Pandas / FinBERT / MiniLM

- Built an end-to-end Python pipeline for regulatory-document processing, semantic similarity, sentiment analysis, preprocessing, validation, and structured financial-text analysis.
- Combined transformer-based representations with similarity and sentiment workflows to support downstream financial research and analytical scoring.

EDUCATION

Bachelor of Technology in Computer Science & Engineering

Dec 2020 – Jul 2024

Sri Aurobindo Institute of Technology, Indore

CGPA: 8.24/10

Eng. Department Rank: 1st

CERTIFICATIONS

- **Python Developer Certification** — MSME, Government of India
- **Crash Course on Python** — Google / Coursera
- **Artificial Intelligence Certification** — Great Learning
- **Prompt Engineering** — DeepLearning.AI

PROFESSIONAL LEARNING

AWS Cloud Solutions Architect

Currently Learning

LEADERSHIP

IEEE Student Branch Chief Secretary

Core Organizing Committee Member — IEEE ICTBIG 2025 and IDBA-ACMWIR 2023

TEDxSAIT 2023

Event Organizing Team Member